

Quick Check

Lesson 8: Nitrogen and Carbon Cycles: How Nutrients Move and Why Algae Boom in Our River

10 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Ecology | Lesson 8

Farmers in Kisii and Bungoma often plant beans between rows of maize. Which process in the bean roots helps the maize crop get more nitrogen?

- A** Bean roots take in nitrogen gas and release it directly to the maize roots
- B** Rhizobium bacteria in root nodules change nitrogen gas from the air into nitrogen compounds
- C** Denitrifying bacteria in the soil change nitrates into nitrogen gas for the maize
- D** Bean leaves use photosynthesis to make nitrates for the maize

A student says, 'Maize plants take nitrogen gas straight from the air and use it to make proteins.' Which is the best correction?

- A** Maize can only get nitrogen by eating dead animals
- B** Maize gets nitrogen from carbon dioxide during photosynthesis
- C** Maize absorbs nitrates from the soil, which were made by nitrogen fixation and nitrification
- D** Maize absorbs nitrogen gas through its stomata, but only at night

A dairy cow in Kericho dies and its body decays in the soil. Which organisms release the nitrogen from its proteins as ammonium so that it can re-enter the nitrogen cycle?

- A** Producers such as maize
- B** Nitrogen-fixing bacteria in bean roots only
- C** Consumers such as other cattle
- D** Decomposers such as bacteria and fungi

A learner eats ugali and then runs. Which process returns carbon from the ugali to the air as carbon dioxide?

- A** Denitrification in the learner's muscles
- B** Nitrification in the learner's stomach
- C** Photosynthesis in the learner's cells
- D** Respiration in the body's cells

Which list contains only processes that return carbon dioxide to the air in the carbon cycle?

A Photosynthesis, burning charcoal and nitrification

B Nitrogen fixation, respiration and burning firewood

C Photosynthesis, respiration and decomposition

D Respiration, decomposition and burning firewood

Rain washes fertiliser from maize farms, cattle waste and sewage into a river. Why does this make algae and water weeds grow very fast?

- A** The fertiliser makes the water colder, so the algae grow faster
- B** The sewage gives the algae more oxygen to breathe
- C** The water gains extra nitrates and phosphates that the algae and weeds use to grow
- D** The algae feed directly on the fish in the river

In a river with an algal bloom, the algae and weeds die in large numbers. Why do fish then die?

- A** The fish die of old age because the water is warmer
- B** The fish have no food because algae are the only food in rivers
- C** Decomposing bacteria multiply and use up dissolved oxygen, so the fish cannot get enough
- D** The dead algae release extra oxygen that is poisonous to fish

Which is the correct order of events in the river after fertiliser and sewage runoff enter it?

- A** Decomposers use up oxygen, nutrient enrichment, algae and weeds die, algal bloom, fish die
- B** Algal bloom, nutrient enrichment, fish die, decomposers use up oxygen, algae and weeds die
- C** Fish die, decomposers use up oxygen, nutrient enrichment, algal bloom, algae and weeds die
- D** Nutrient enrichment, algal bloom, algae and weeds die, decomposers use up oxygen, fish die

Along a polluted river, people notice a foul smell after many algae and weeds have died. What is the best explanation?

- A** The smell comes from nitrogen gas escaping from the water
- B** With little oxygen left, anaerobic decay of the dead matter produces foul-smelling substances
- C** Fish release a bad smell when they are hungry
- D** Algae produce a bad smell only when they are alive and healthy

A student hears a group claim 'the fish are disappearing because they have no food.' What evidence from the river would best show this claim is incomplete?

- A** The river still has some fish in it
- B** The river is flowing towards Lake Victoria
- C** Algae and weeds are growing fast while dissolved oxygen falls after they die and decompose
- D** Beans are planted near the river

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).